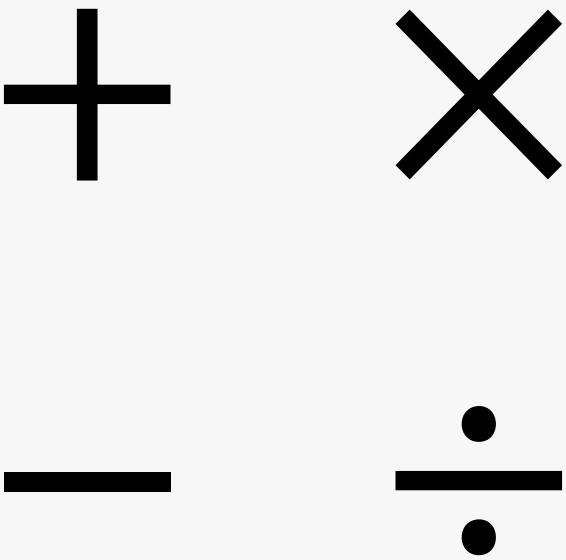


29.09.26

Polynomial is formal expression

Proposition
A (non-zero) polynomial of degree $n \geq 0$,
i.e. $P(X) = a_n X^n + a_{n-1} X^{n-1} + \dots + a_1 X + a_0$
with $a_n \neq 0$,
can't have more than n roots

Corollary: let $g(X) \in \mathbb{Q}[X]$ polynomial,
$\deg(g) \leq d$, and let $X_0, X_1, \dots, X_d \in \mathbb{Q}$ pairwise
distinct rational values. Set
$y_i := g(x_i)$ for $i =$

If $h(X) \in \mathbb{Q}[X]$ with $\deg(h) \leq d$ and $h(x_i) = y_i$, then $g(X) = h(X)$

Proof: Assume towards a contradiction
that a h like above exists but for $g \neq h$.

Then $0 \neq P$

Encoding a message

Message: 3 numbers
a_0, a_1, a_2 (eg. representing letters of a text)

encode message as a polynomial

$f(x) = a_0 + a_1 x + a_2 x^2$

choose 5 eval points

$x_i = 0, 1, 2, 3, 4$

Compute $f(0), f(1), f(2), f(3), f(4)$

These 5 numbers comprise the codeword that is transmitted.

Redundancy used to protect message against errors during transmission. (eg. noise)

Example:
$a_0 = 2, \ a_1 = 1, \ a_2 = 3$

$f(x) = 2 + x + 3x^2$

$f(0) = 2$ $\quad$ $f(3) = 32$
$f(1) = 6$ $\quad$ $f(4) = 54$
$f(2) = 16$

message we transmit: 2 6 16 32 54

Any 3 correct values
$\Rightarrow$ original message.

For example,

$f(0)=2,\ f(1)=6,\ f(3)=32$
need to find $a_0, a_1, a_2 \in \mathbb{R}$ for

$f(x)=a_0+a_1x+a_2x^2$ s.t.

$f(0)=a_0+0a_1+0a_2=2$
$f(1)=a_0+a_1+a_2=6$
$f(3)=a_0+3a_1+9a_2=32$

Solve linear system using aug. matrix.

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 1 & 1 & 1 & 6 \\ 1 & 3 & 9 & 32 \end{array}\right) \quad \text{e.r.o}$$

$$\Rightarrow \left(\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 1 & 4 \\ 0 & 3 & 9 & 30 \end{array}\right) \Rightarrow \left(\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 1 & 4 \\ 0 & 1 & 3 & 10 \end{array}\right) \Rightarrow \left(\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 1 & 4 \\ 0 & 0 & 2 & 6 \end{array}\right)$$

$$\Rightarrow \left(\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 1 & 4 \\ 0 & 0 & 1 & 3 \end{array}\right) \Rightarrow \left(\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 3 \end{array}\right)$$

a_0, a_1, a_2

$f(x)=2+x+3x^2$

Using error of 99, $f(99)\neq 0$, $\therefore$ error somewhere.

MATH3011 Coding theory and cryptography

1.2.2 Quadratic Polynomials

$f(x) = ax^2 + bx + c = 0, \quad a \neq 0$

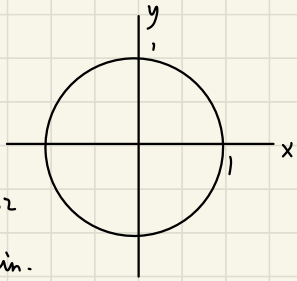
$\rightsquigarrow x_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

No G.S. for degree ≥ 5. Galois theory.

Studying solutions of quadratic polynomial equations
leads to irrational numbers ($\sqrt{2}$), complex numbers ($\sqrt{-1} = i$)

Look at $x^2 + y^2 = 1$

$\{(x,y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}$



describes all points (x,y) in $\mathbb{R}^2$
that are distance 1 from origin.

What are all rational solutions for $x^2 + y^2 = 1 \quad x, y > 0$ and $\in \mathbb{Q}$

Write $x = \frac{a}{c}$, $y = \frac{b}{c}$ for $a, b, c \in \mathbb{Z}$

$\gcd(a,b,c) = 1$

$\frac{a^2}{c^2} + \frac{b^2}{c^2} = 1 \Rightarrow a^2 + b^2 = c^2$

Primitive/reduced Pythagorean triples

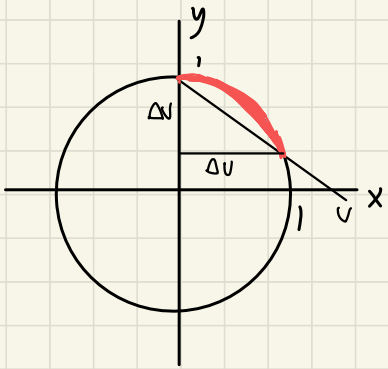
If $\gcd(a,b,c) \neq 1$ then it is not primitive.

$\left(\frac{a}{\gcd(a,b,c)}, \frac{b}{\gcd(a,b,c)}, \frac{c}{\gcd(a,b,c)} \right)$ will give primitive triple.

$(3,4,5)$ is primitive Pythagorean triple.
any multiple is Pythagorean triple.

01.10.26

How to get all primitive Pythagorean triples?



we get a bijection

Positive arc

$\{(X,y) \subseteq \mathbb{R}^2 : x^2+y^2=1\} \overset{1:1}{\longleftrightarrow} (1,+\infty)$

α: $\rightarrow$

β: $\leftarrow$

Let's determine the map α

Line through $(0,1)$ and (x,y)

$V = \frac{y-1}{x} \cdot v + 1$

Intersection with X axis:

$0 = \frac{y-1}{x} \cdot v + 1 \quad \Rightarrow V = \frac{x}{1-y} \quad \Rightarrow \alpha(x,y) = \frac{x}{1-y}$

We get map:

$$\alpha : \{(x,y) \in \mathbb{Q}^2 > 0 : x^2+y^2=1\} \to \mathbb{Q}>1$$

$$(x,y) \mapsto \frac{x}{1-y}$$

Now compute beta.

Let $u \in \mathbb{Q}>1$

Line through $(0,1)$ and $(u,0)$

$$y = \frac{-1}{u} \cdot x + 1$$

$$= \frac{x-u}{-u}$$

Insert into circle

$$x^2+y^2=1$$

$$x^2+\left(\frac{x-u}{-u}\right)^2 = 1$$

$$x^2+\frac{(x-u)^2}{u^2} = 1$$

$$u^2x^2+x^2-2ux+u^2 = u^2$$

$$(u^2+1)x^2-2ux = 1$$

$\Rightarrow x = 0 \quad \text{or} \quad x = \frac{2v}{v^2+1}$

So we want

$y = \frac{-1}{v}\left(\frac{2v}{v^2+1}\right) + 1 = \frac{v^2-1}{v^2+1}$

$\Rightarrow \beta(v) = (x,y) = \left(\frac{2v}{v^2+1}, \frac{v^2-1}{v^2+1}\right)$

rational if v is rational

We get bijection

$\mathbb{Q} > 1 \to \{(x,y) \in \mathbb{Q}^2 > 0 : x^2 + y^2 = 1\}$

Parameterise

$v = \frac{m}{n} \qquad \gcd(m,n) = 1$

then $(x,y) = \left(\frac{2(m/n)}{(m/n)^2+1}, \frac{(m/n)^2-1}{(m/n)^2+1}\right)$

$\Rightarrow \left(\frac{2mn}{m^2+n^2}, \frac{m^2-n^2}{m^2+n^2}\right)$

$\Rightarrow (a,b,c) = (2mn, m^2-n^2, m^2+n^2)$

not necessarily primitive.

Eg. $m=3, n=1$

$(a,b,c)=(6,8,10)$ not primitive

1.2.3 Counting and Power Series

Q: How many ways can you make 20p with 1p, 2p, 5p, 10p, 20p?

Power series

$I=(a,b)\subseteq\mathbb{R}$, $I=\mathbb{R}$

$f(x)=\sum_{n=0}^{\infty} f_n x^n \quad (f_n\in\mathbb{R}, f_n\in\mathbb{C})$ is a power series

$f_n=\frac{f^{(n)}(0)}{n!}$

Eg. $f(x)=1+x+x^2+\cdots$

$=\sum_{n=0}^{\infty} x^n$ geometric analysis

Example: Find power series of

$f(x)=\frac{1}{(1-x)^2}$ Well $\frac{d}{dx}\left(\frac{1}{1-x}\right)=(-1)\cdot(1-x)^{-2}\cdot(-1)=\frac{1}{(1-x)^2}$

$f(x)=\frac{d}{dx}\left(\frac{1}{1-x}\right)=\frac{d}{dx}\sum_{n=0}^{\infty}x^n=\sum\frac{d}{dx}x^n=\sum_{n=1} nx^{n-1}$ let $m=n-1 \Rightarrow \sum_{m=0}^{\infty} m$

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Example: How many ways make 20p using 1,2,5,10,20p?

only 1p's

$c_n :=$ number of way to make n'p with 1p's $= 1$

Use as co-efficients of power series

$$\rightsquigarrow \underset{c_0}{1}\cdot(x^1)^0 + \underset{c_1}{1}(x^1)^1 + \underset{c_2}{1}(x^1)^2 + \cdots$$

$$= \sum_{n=0}^{\infty} x^n$$

for 1p, 2p for $n \in \mathbb{Z}_{\geq 0}$ using 1p's and 2p's

first look at only 2p's

d_n: number of ways to make n'p with 2p's

$$= \begin{cases} 0 & n \text{ odd} \\ 1 & n \text{ even} \end{cases}$$

generating series $1\cdot x^0 + 0\cdot x^1 + 1\cdot x^2 + 0\cdot x^3 + \cdots$

$$= \sum_{n=0}^{\infty} x^{2n}$$

To get # ways to make n'P from 1p and 2p, we multiply geom series

$$\left((x^1)^0 + (x^1)^1 + (x^1)^2 + \cdots\right)\left((x^2)^0 + (x^2)^1 + (x^2)^2 + \cdots\right)$$

$$= \frac{1}{1-x}\cdot\frac{1}{1-x^2} = \sum_{n=0}^{\infty} e_n x^n$$

$$\frac{1}{1-x} \cdot \frac{1}{1-x^2} = \sum_{n=0}^{\infty} e_n x^n$$

For $n=5$ using 1p, 2p

$$(1 + x + x^2 + x^3 + x^4 + x^5 + \cdots)(1 + x^2 + x^4 + \cdots)$$

$$= x^5 \cdot 1 + x^3 \cdot x^2 + x^1 \cdot x^4 + \cdots$$

$$= 3 \cdot x^5 + \cdots$$

So 3 ways to make 5p from 1p, 2p

Back to original question

$$\underset{1p}{(1 + x + x^2 + \cdots)} + \underset{2p}{(1 + x^2 + x^4 + \cdots)} + \underset{5p}{(1 + x^5 + x^{10} + \cdots)} + \underset{10p}{(1 + x^{10} + x^{20} + \cdots)}$$

$$= \frac{1}{1-x} \cdot \frac{1}{1-x^2} \cdot \frac{1}{1-x^5} \cdot \frac{1}{1-x^{10}}$$

$$= \sum_{n=0}^{\infty} a_n x^n$$

each way to make n corresponds uniquely to term a_n in power series
$\rightsquigarrow$ each way to make n contributes one to coefficient of x^n

for example $(x^1)^{11} + (x^2)^2 + (x^5)^1 + (x^{10})^0 = x^{20}$ is one way

Using Python for $n=20$, there are 40 ways.

Store sells:

- Shirts £5
- Pants £10
- Shoes £25

Shirts generating series

$$1 + x^5 + x^{10} + \cdots = \sum_{n=0}^{\infty} x^{5n} = \frac{1}{1-x^5}$$

pants

$$1 + x^{10} + x^{20} + \cdots = \sum_{n=0}^{\infty} x^{10n} = \frac{1}{1-x^{10}}$$

Shoes

$$1 + x^{25} + x^{50} + \cdots = \sum_{n=0}^{\infty} x^{25n} = \frac{1}{1-x^{25}}$$

For all we take product

Spend £50

$$\left(\frac{1}{1-x^5}\right) \cdot \left(\frac{1}{1-x^{10}}\right) \cdot \left(\frac{1}{1-x^{25}}\right)$$

→ Use computer algebra = 10

What about spend at most £50

We want x^n for $n \leq 50$

$$T(x) = 10x^{50} + 8x^{45} + 7x^{40} + 6x^{35} + 5x^{30} + 4x^{25} + 3x^{20} + 2x^{15} + 2x^{10} + \cdots$$

$$= 49$$

Proposition

1 3 Diophantine equations

Fermat's Last theorem
For fixed $n \geq 3$, the only integer solutions to $x^n + y^n = z^n$, satisfy $xyz = 0$

(Proved by Andrew Wiles in 1990's)

Proof: Very difficult!

Diophantine equation
An equation where

Example
$x^2 - 2y^2 = 0$

$(x, y) = (0, 0)$ only integer solution

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- If "a" appears must appear 4 times
- "β" appears even times
- "c" at least 3 times
- "d" no more than 5 times

a : $1 + x^4$
(1 : a not used, x^4 : used 4 times)

b : $1 + x^2 + x^4 + \cdots = \frac{1}{1-x^2}$

c : $x^3 + x^4 + x^5 + \cdots = \frac{x^3}{1-x}$

d : $1 + x + x^2 + \cdots + x^5$

For 10 letters

$$(1+x^4)\frac{1}{1-x^2}\left(\frac{x^3}{1-x}\right)(1+x+\cdots+x^5)$$

⇝ Use computer algebra coefficient of x^{10}

→ 24 unordered words of length 10 satisfying restrictions.

Example: Find number of integer solutions to

$$x_1 + x_2 + x_3 + x_4 = 15$$

s.t. $0 \le x_1 \le 10$

$0 \le x_2 \le 5$

$4 \le x_3 \le 7$

$2 \le x_4 \le 8$

$A_1(x), A_2(x), A_3(x), A_4(x)$

exponent in $A_i(x)$ gives value of x_i

$A_1(x) = 1 + x + \cdots + x^{10}$

$A_2(x) = 1 + x + \cdots x^5$

$A_3(x) = x^4 + \cdots x^7$

$A_4(x) = x^2 + \cdots + x^8$

$\rightsquigarrow$ Compute $A_1(x)A_2(x)A_3(x)A_4(x)$

Coefficient of $x^{15} = 134$

Question: Find all $a, b \in \mathbb{N}$ s.t. $a^2 + b^2 = 65^2$

Use parameterisation

$(2mn, m^2 - n^2, m^2 + n^2)$

$m^2 + n^2 = 65$

M	N
2	1
3	2, 1
4	3, 2, 1
5	4, 3, 2, 1
6	5, 4, 3, 2, 1
7	6, 5, 4, 3, 2, 1
8	7, 6, 5, 4, 3, 2, 1

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First 20 mins

Example: Show that $x^3+x+1=0$ has no integer roots

Rational root test gives candidate solutions, divisors of $a_0=1$

$\rightsquigarrow x_1=-1, x_2=1$

Test candidates

$$(-1)^3+(-1)+1=-1\neq 0$$

$$(1)^3+(1)+1=3\neq 0$$

$\Rightarrow$ eq, has no integer solutions. $\square$

Proposition: Reduction modulo n criterion.
If $\star$ has no solutions modulo an integer $m>1$, then it has no solutions over $\mathbb{Z}$.

Example: Show $11x^3+77x^2+66x^3+1=0$ (1)
has no integer solutions

Taking (1) modulo m is equivalent to thinking of it as an equation over $(\mathbb{Z}/m\mathbb{Z},+)$

Use reduction mod $m=11$,

(2) $\quad 1\equiv 0 \pmod{11}$

(2) eq, has no solution
$\Rightarrow$ eq, (1) has no solution

Proof of reduction criterion

If $*$ has no sol. over $\mathbb{Z}$, then $*$ mod m $\forall m > 1$ has a sol.

Second Statement

Theorem (Eisenstein criterion):

Suppose $n > 1$,

Suppose $\exists$ a prime p s.t.

- $p \nmid a_n$
- $p \mid a_i$ for $0 \le i \le n-1$
- $p^2 \nmid a_0$

Then $*$ has no integer solutions.

Example: Show that

$$x^4 + 5x^3 + 10x^2 + 10x + 5 = 0$$

has no int. solutions

try $p = 5$: $5 \nmid 1$

$5 \mid 5$

$5 \mid 10$

$5 \mid 10$

$25 \nmid 5$

All assumptions of Eisenstein satisfied $\therefore$

no integer sols.

Exercise: Let $n \in \mathbb{Z}$, $n > 1$.

Show $x^n + 5x^{n-1} + \cdots + 5x + 5 = 0$

and

$x^n + 5 = 0$

has no integer solutions.

Proof Eisenstein's criterion

Linear Diophantine Equations

* $ax + by = c$

for $a, b, c \in \mathbb{Z}$

$a \neq 0, b \neq 0, ab \neq 0$

Example: Solve $15x+37y=22$

Sol: we note $d=(15,37)=1$

Find $\xi, \eta \in \mathbb{Z}$ s.t.

$15\cdot\xi+37\eta=1$

Euclidean algorithm

$37=\underbrace{15}_{a}\cdot 2+\underbrace{7}_{r_1} \iff 7=37-15\cdot 2$

$15=2\cdot 7+\underbrace{1}_{(a,b)=d}$

$1=15-2\cdot 7$

$=15-2\cdot(37-15\cdot 2)$

$=15-2\cdot 37+4\cdot 15$

$=\underbrace{5}_{\xi}\cdot 15-\underbrace{2}_{\eta}\cdot 37$

Theorem above gives all solutions:

$(x,y)=(5\cdot 22+37\cdot k,(-2)\cdot 22-15\cdot k)$

$=(110+37k,-44-15k)$ for $k\in\mathbb{Z}$

Remark

Sometimes you can spot solution

(e.g. $(-1,1)$ in above), say (x_0,y_0)

Full sol is $(x_0+b\cdot k, y_0-a\cdot k)$ $\forall\, k\in\mathbb{Z}$

Coursework 1 (Due 16.10.26)

Let $a(n)$ denote the number of ways of distributing n identical sweets between Alice and Charlie, subject to the conditions that Alice receives at least two sweets and Charlie receives an even number of sweets.

1. (a) Express the generating function

$$A(x) = \sum_{n \geq 0} a(n) x^n$$

as a fraction of two polynomials. Justify your answer. You should explain how each factor in your generating function arises. A correct generating function without justification will receive a maximum of 5 marks. [40 marks]

First we need to construct a generating function for Alice and Charlie seperately.
The exponent of each term x^n equals the number of sweets for that person.
The coefficient of x^n is how many ways to make n sweets.

Alice's generating function $B(x)$:

· At least two sweets

$$B(x) = x^2 + x^3 + x^4 + x^5 + \cdots = x^2(1 + x + x^2 + x^3 + \cdots)$$

$$1 + x + x^2 + x^3 + \cdots = \sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$$

$$\therefore B(x) = x^2 \left(\frac{1}{1-x}\right) = \frac{x^2}{1-x}$$

Charlie's generating function $C(x)$:

· even number of sweets

$$C(x) = 1 + x^2 + x^4 + x^6 + \cdots = \sum_{n=0}^{\infty} x^{2n} = \frac{1}{1-x^2}$$

$$A(x) = B(x)C(x) = \left(\frac{x^2}{1-x}\right) \cdot \left(\frac{1}{1-x^2}\right) = \frac{x^2}{(1-x)(1-x^2)}$$

(b) Using partial fractions applied to the generating function you obtained in part (a), find an explicit formula for $a(n)$, valid for all $n \geq 0$. Hence, verify that $a(51) = 25$. Justify your answer. [60 marks]

6) $A(x) = \dfrac{x^2}{(1-x)(1-x^2)}$

Notice $(1-x^2)$ is the difference of two squares.

$1-x^2 = (1-x)(1+x)$

So $A(x) = \dfrac{x^2}{(1-x)^2(1+x)} = \dfrac{A}{(1-x)^2} + \dfrac{B}{1-x} + \dfrac{C}{1+x}$

Multiplying by common denominator

$x^2 = A(1+x) + B(1-x)(1+x) + C(1-x)^2$

let $x=1$: $1^2 = 2A + 0B + 0C \Rightarrow A = \frac{1}{2}$

let $x=-1$: $(-1)^2 = 0A + 0B + (2^2)C \Rightarrow C = \frac{1}{4}$

let $x=0$, and plug in A and C that we found:

$0^2 = \frac{1}{2}(1) + B(1)(1) + \frac{1}{4}(1)^2$

$\Rightarrow B = -\frac{1}{2} - \frac{1}{4} = -\frac{3}{4}$

$\therefore A(x) = \dfrac{1}{2(1-x)^2} - \dfrac{3}{4(1-x)} + \dfrac{1}{4(1+x)}$

$$A(x) = \frac{1}{2(1-x)^2} - \frac{3}{4(1-x)} + \frac{1}{4(1+x)}$$

We want $A(x)$ in the form $A(x) = \sum_{n \geq 0} a(n) x^n$

First rewrite $A(x)$ as a series

$$A(x) = \frac{1}{2} \cdot \left(\frac{1}{(1-x)^2} \right) - \frac{3}{4} \cdot \left(\frac{1}{1-x} \right) + \frac{1}{4} \cdot \left(\frac{1}{1+x} \right)$$

We know

$$\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$$

taking the derivative w.r.t x

$$\frac{d}{dx} \left(\sum_{n=0}^{\infty} x^n \right) = \frac{d}{dx} \left(\frac{1}{1-x} \right)$$

LHS: $\sum_{n=1}^{\infty} n x^{n-1}$

let $m = n-1$

$$\Rightarrow \sum_{m=0}^{\infty} (m+1) x^m$$

RHS: Using Chain rule

$$\frac{d}{dx}(1-x)^{-1} = (-1)(1-x)^{-2}(-1) = \frac{1}{(1-x)^2}$$

$$\therefore \frac{1}{(1-x)^2} = \sum_{n=0}^{\infty} (n+1) x^n$$

$$\frac{1}{1+x} = \frac{1}{1-(-x)} = \sum_{n=0}^{\infty}(-x)^n = \sum_{n=0}^{\infty}(-1)^n x^n$$

Combining these results gives

$$A(x) = \frac{1}{2}\sum_{n=0}^{\infty}(n+1)x^n - \frac{3}{4}\sum_{n=0}^{\infty}x^n + \frac{1}{4}\sum_{n=0}^{\infty}(-1)^n x^n$$

Collecting like terms

$$A(x) = \sum_{n=0}^{\infty}\left[\left(\frac{1}{2}(n+1) - \frac{3}{4} + \frac{1}{4}(-1)^n\right)x^n\right]$$

Simplifying

$$A(x) = \sum_{n=0}^{\infty}\left[\underbrace{\left(\frac{1}{2}n + \frac{1}{2} - \frac{3}{4} + \frac{1}{4}(-1)^n\right)}_{a(n)}x^n\right]$$

this gives $a(n) = \frac{1}{2}n - \frac{1}{4} + \frac{1}{4}(-1)^n$

in $A(x) = \sum_{n=0}^{\infty} a(n)\,x^n$

Verifying $a(51)$

$$a(51) = \frac{1}{2}(51) - \frac{1}{4} + \frac{1}{4}(-1)^{51}$$

$$(-1)^n = \begin{cases} 1 & n = \text{even} \\ -1 & n = \text{odd} \end{cases}$$

$a(51) = \frac{51}{2} - \frac{1}{4} - \frac{1}{4} = \frac{51}{2} - \frac{1}{2} = \frac{50}{2} = 25$ as required.

15.10.26

Want to solve $x^{2k} + x + 1 = 0$
over the integers where $k \in \mathbb{Z}_{\geq 1}$

We have 3 tests
- Eisenstein criterion ✗ coefficients all 1
- reduction modulo n
- rational root test

rational root test

Candidate solutions are $x = 1$ or $x = -1$

$(-1)^{2k} + (-1) + 1 = 1 \neq 0$
$(1)^{2k} + (1) + 1 = 3 \neq 0$

$\Rightarrow$ Eq has no sol over $\mathbb{Z}$

b) Using reduction mod $m = 2$

$(0)^{2k} + (0) + 1 = 1 \neq 0$
$(1)^{2k} + (1) + 1 = 3 \equiv 1 \bmod 2 \neq 0$

$\Rightarrow$ eq has no solutions mod 2 $\therefore$ by reduction criterion
no sols over $\mathbb{Z}$

Solve $x^6+9x^4+6x^2+3=0$ *

Eisenstein for $p=3$

$3 \nmid 1$
$3 \mid 9$
$3 \mid 6$
$9 \nmid 3$

$\therefore$ By Eisenstein, no solutions over $\mathbb{Z}$.

Solve $18x+30y=222$
over integers

$\gcd(18,30,222)=6$

$3x+5y=37$

Can spot $(-1,8)$

Using remark: $(x,y)=(-1+5k, 8-3k)$
x_0 b y_0 $-a$

Addition: we want $x \geq 0, y \geq 0$

$-1+5k \geq 0 \Rightarrow 5k \geq 1 \Rightarrow k \geq \frac{1}{5}$
$8-3k \geq 0 \Rightarrow 8 \geq 3k \Rightarrow k \leq \frac{8}{3}$

$\frac{1}{5} \leq k \leq \frac{8}{3}$ $\therefore k=1$ or 2
$\Rightarrow (4,5), (9,2)$

Show $x^2+y^2+z^2 = 7w^2$
has no integer sols other than $(0,0,0,0)$

Remark
- $a^2 \equiv 0,1 \bmod 4 \quad \forall k \in \mathbb{Z}$
- $(2k+1)^2 \equiv 1 \bmod 8$ " "

Because $(2k+1)^2 = 4k^2+4k+1$
$= 4(k+1)k+1$
one of these is even.

let $(x,y,z,w) \in \mathbb{Z}^2$ be a solution

Case 1 $2 \nmid w$

reduce modulo 4

$x^2+y^2+z^2 = 7w^2$

mod 4
$x^2+y^2+z^2 = 3w^2$

Case 2: $2 \mid w$

Assume towards a contradiction

Choose solution where $|w|$ is minimal

$|\&|w| = 0$ then $x, y, z = 0$

For $|w| > 0$, take modulo 4

$\Rightarrow x^2 + y^2 + z^2 \equiv 0 \bmod 4$

$\Rightarrow 2 \mid x, 2 \mid y, 2 \mid z$



2020.2.6

Proof

Case $d>1$:

Assume $\exists\ x,y \in \mathbb{Z}$ s.t.

$ax+by=c$

$d=(a,b) \Rightarrow d|a$ and $d|b$

Hence $d|(ax+by)$

$\Rightarrow d|c=ax+by$

However, we have primitive eq., that is, $(c,d)=1$ ↯

Case $d=1$:

Reduce equation $ax+by=c \mod b$

That is, we look for solutions of

$ax=c \pmod b$

Generalisation: more than 2 variables

$$a_1 x_1 + a_2 x_2 + \cdots + a_n x_n = c$$

$a_1, \ldots, a_n, c \in \mathbb{Z}$

with $d = (a_1, \ldots, a_n)$
$(c, d) = 1$

one route

1. Solve
   $\delta \cdot y + a_n x_n = c$
   where $\delta = (a_1, \ldots, a_{n-1})$

2. Solve
   $$\frac{a_1}{\delta} x_1 + \cdots + \frac{a_{n-1}}{\delta} x_{n-1} = y$$

Non-linear Diophantine equations in multiple variables



For example

$x^4+17x^2y^6+9z^2+19=0$

Lemma 1.3.12:

Let $f(x,y,z)$ be polynomial with integer co-efficients

If $f(x,y,z)>0$ $\forall$ real x,y,z then $f(x,y,z)=0$ has no integer solution.

Example:

$x^4+17x^2y^6+9z^2+19=0 \quad \forall x,y,z \in \mathbb{Z}$

$x^4 \geq 0, x^2y^6 \geq 0, z^2 \geq 0$

So $x^4+17x^2y^6+9z^2+19 \geq 0$

$\therefore$ eq, has no integer solutions

Example

Solve $x^2+y^2+2y+1=0$

$x^2+(y^2+2y+1)=0$

$x^2+(y+1)^2=0$

$\Rightarrow x=0$ and $y+1=0$

$\Rightarrow y=-1$

Unique solution $x=0, y=-1$ $\square$

recap of modulo arithmetic

Fix $k \in \mathbb{N}$, $m, n \in \mathbb{Z}$ are congruent modulo k if $k \mid (m-n)$

write $m = k \bmod n$

"$= \bmod k$" equivalence class

$[x]_k = [x]_{\equiv k}$

$[x] = \bar{x}$

well defined operations

$+ : \mathbb{Z}/k\mathbb{Z} \times \mathbb{Z}/k\mathbb{Z} \to \mathbb{Z}/k\mathbb{Z}$

$(m+k\mathbb{Z}) \quad (n+k\mathbb{Z}) \quad ((m+n)+k\mathbb{Z})$

$\times : \mathbb{Z}/k\mathbb{Z} \times \mathbb{Z}/k\mathbb{Z} \to \mathbb{Z}/k\mathbb{Z}$

$mk\mathbb{Z} \quad nk\mathbb{Z} \quad (mn)k\mathbb{Z}$

Lemma 1.3.16 (reduction criterion)

let $S(x,y,z,\ldots)$ be a polynomial with integer co-efficients.

If $S(x,y,z,\ldots) \equiv 0 \bmod m$ has no solutions

Example:

$x^3 - y^3 = 3$ over integers

Take modulo 9

x	0	1	2	3	4	5	6	7	8
$x^3 \bmod 9$	0	1	-1	0	1	-1	0	1	-1

$$4^3 = 4^2 \cdot 4 = \overline{16} \cdot 4 = 7 \cdot 4 = \overline{28} = 1$$

$\Rightarrow x^3 - y^3 = \{-2,-1,0,1,2\} \bmod 9$

$\Rightarrow x^3 - y^3 \neq 3 \bmod 9$

$\rightarrow$ by lemma, no integer solution.

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Worksheet 1

1. Eisenstein Criterion

a) Theorem (Eisenstein criterion):
Suppose $n>1$,
Suppose $\exists$ a prime p s.t.

- $p \nmid a_n$
- $p \mid a_i$ for $0 \le i \le n-1$
- $p^2 \nmid a_0$

Then * has no integer solutions.

b) $x^4+5x^3+10x^2+10x+5=0$, using Eisenstein

Taking $p=5$,

$5 \nmid 1$
$5 \mid 5$
$5 \mid 10$
$5 \mid 10$
$5^2=25 \nmid 5$

$\therefore$ by Eisenstein Criterion, equation has no integer solutions

c) $x^4+9x^3+31x^2+49x+31=0$

Use $y=x+1$
$\Rightarrow x=y-1$

$(y-1)^4+9(y-1)^3+31(y-1)^2+49(y-1)+31=0$

$(y-1)^4+9(y-1)^3+31(y-1)^2+49(y-1)+31=0$

$y^4+4y^3(-1)+6y^2(-1)^2+4y(-1)^3+(-1)^4=y^4-4y^3+6y^2-4y+1$

$y^3+3y^2(-1)+3y(-1)^2+(-1)^3=y^3-3y^2+3y-1 \quad \cdot(9)=9y^3-27y^2+27y-9$

$y^2+2y(-1)+(-1)^2=y^2-2y+1 \quad \cdot(31) \quad =31y^2-62y+31$

$49y-49$

$+31$

Whole equation

$y^4+(9y^3-4y^3)+(6y^2-27y^2+31y^2)+(27y-4y+49y-62y)+(1-9+31+31-49)$

$=y^4+5y^3+10y^2+10y+5=0$

Let $P=5$,

$5\nmid 1$
$5\mid 5$
$5\mid 10$
$5\mid 10$
$5^2\nmid 5$

$\therefore$ by Eisenstein Criterion, no integer solutions

d) $x^2+5x-11=0$

$y=x-2$

$x=y+2$

$(y+2)^2+5(y+2)-11=0$

$y^2+4y+4+5y+10-11=0$

$y^2+9y+3=0$

Taking $p=3$ $\quad p \nmid a_0,\ p \mid a_1,\ p^2 \nmid a_2$

$\therefore$ by Eisenstein

2. a) $34x+85y=13$

17
34
51
68
85

Take reduction modulo 17

$0 \equiv 13 \bmod 17$ which has no solutions

$\therefore$ eq has no integer solutions

b) $2y^2+4xy+3x^2+2x+3=0$ *

$(x+y)^2=x^2+2xy+y^2$

$2(x+y)^2=2x^2+4xy+2y^2$

$2(x+y)^2+x^2+2x+3=0$ *

$2(x+y)^2+(x+1)^2+2=0$

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Example: Show for all $x, y \in \mathbb{Z}$

$$x^2 + y^2 \neq 103$$

Use reduction criterion for $m = 4$

x	0	1	2	3
$x^2 \bmod 4$	0	1	0	1

Notice $103 = 3 \bmod 4$

$$x^2 + y^2 \equiv 0, 1, 2 \bmod 4$$

but we need $x^2 + y^2 \equiv 3 \bmod 4$

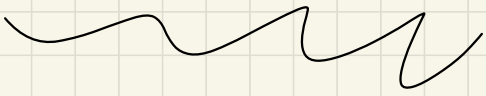
Contradiction

$$\Rightarrow x^2 + y^2 \neq 103 \quad \forall x, y \in \mathbb{Z}$$

Example: Diophantine equation

$$x^4 + y^4 + z^4 = t^4 \quad (x, y, z, t \text{ all non-zero})$$

Smallest solution:
$95800^4 + 217519^4 + 414560^4 =$



1.3.4 Wilson's theorem

Recall $(\mathbb{Z}/k\mathbb{Z}, +)$ group but not multiplication.
$\bar{z} \in \mathbb{Z}/6\mathbb{Z}$

$(\mathbb{Z}/k\mathbb{Z})^\times \subseteq \mathbb{Z}/k\mathbb{Z}$ have multiplicative inverse.

e.g. $(\mathbb{Z}/6\mathbb{Z})^\times = \{\bar{1}, \bar{5}\}$
$= \{\bar{c} : \gcd(c, 6) = 1\}$

$((\mathbb{Z}/k\mathbb{Z})^\times, \cdot)$ always abelian group

Recall Lagrange's thm
$(G, *)$ finite group
let $g \in G$.

$$g^{*|G|} = g * \cdots * g \text{ times} = e$$
$$|G| \text{ times}$$

For a prime p,
$(\mathbb{Z}/p\mathbb{Z})^\times = \{\bar{1}, \bar{2}, \ldots, \overline{p-1}\}$ (abelian group $\bar{1}$ neutral element w.r.t $\cdot$)

Fermat's little theorem
let $p \in \mathbb{N}$ prime, and $a \in \mathbb{Z}$, not divisible by p. Then

$$a^{p-1} \equiv 1 \bmod p$$

Fermat concerns powers of one element.

Conjecture:

If p prime then $(p-1)! \equiv (p-1) \bmod p$

$\equiv -1 \bmod p$

Try Composite numbers:

$m=4 \quad 3! = 6 \equiv 2 \bmod 4$

$m=6 \quad 5! = 120 \equiv 0 \bmod 6$

$m=8 \quad 7! = 5040 \equiv 0 \bmod 8$

$m=9 \quad 8! = 40320 \equiv 0 \bmod 9$

Conjecture: If m composite number, then

$$(m-1)! = \begin{cases} 2 \bmod 4 & \text{if } m=4 \\ 0 \bmod m & \text{if } m>4. \end{cases}$$

Thm (Wilson)

$m \in \mathbb{N}$ prime $\Leftrightarrow (m-1)! \equiv -1 \bmod m$

Proof:

$\Rightarrow$ Suppose $m=p$ prime

Claim: let $\bar{a} \in (\mathbb{Z}/m\mathbb{Z})^\times$ then

$\bar{a}^{-1} \neq \bar{a} \Leftrightarrow \bar{a} \neq \bar{1}, \overline{-1}$

Example